

Sriram Hariharan Neelakantan

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PROFESSIONAL EXPERIENCE

Techmentee Inc. , Sr. Data Scientist | Harrison, NJ

Jul 2025 – Current

- Cut average response time from 3 min to 45 sec by developing an LLM-based multi-agent intent framework (LangGraph + Step Functions) automating tier-1 FinTech chats like balance and KYC queries
- Expanded automation to nine workflows by building API-driven Action Agents (Lambda + Step Functions) for refund, payment retry, and card block, removing human review for low-risk cases
- Boosted resolution accuracy by 12% via a human-in-the-loop feedback loop retraining classifier agents and refining Bedrock prompt logic with dynamic evaluation metrics

L.B. Foster Salient Systems, Data Scientist II | Dublin, OH

Dec 2024 – Jun 2025

- Built scalable data pipelines using Python & PySpark on Databricks with Azure Storage to process 3M+ high-speed sensor records, enabling automated wheel filtering, anomaly detection, poly-curve analysis, and weight calculation
- Developed a wheel health metric using double differential filtering and polynomial curve fitting with K-means, filtering out 35% of jittery wheels and detecting 18% more true electronic anomalies
- Partnered with railway clients, operations teams, and engineering leads to deliver Power BI dashboards backed by SQL queries from Databricks, improving maintenance KPI visibility and planning accuracy
- Automated ADC-based sensor data validation across 16 circuits per rail, reducing misclassification by 40% and strengthening anomaly isolation and wheel health diagnostics
- Performed advanced EDA in Python and PySpark on Databricks to identify trends and anomalies, enabling proactive wheel replacement and optimized maintenance schedules

Norfolk Southern Corp, Machine Learning Intern | Atlanta, GA

Sep 2022 – Dec 2022

- Applied KNN, PCA, and K-means clustering to enhance spatial analysis of 90k+ rail-road intersections, improving infrastructure planning accuracy
- Streamlined EDA using Pandas, NumPy, and Matplotlib to surface 50+ anomalies across 30k+ records, increasing clustering performance by 40%
- Implemented Faster R-CNN in PyTorch to detect defects in railway components, classifying 1,000+ defects from 7k undercarriage images
- Improved defect classification accuracy by 12% through advanced image processing techniques including edge detection and segmentation

Accenture, Data Analyst II | India

Mar 2020 – Jul 2021

- Architected ML-driven data pipelines using Apache Spark for insurance campaign response analysis, improving conversion rates by 5.5%
- Deployed gradient boosting and logistic regression to segment high-risk customers, reducing early churn by 15% through targeted retention strategies
- Automated ingestion and transformation of 250k+ customer records weekly using SQL and Python, achieving 98% data quality
- Launched Tableau dashboards consolidating retention and engagement KPIs, shortening marketing decision cycles by 30%

- Designed fraud detection models using gradient boosting and anomaly detection, identifying 22% more suspicious claims and mitigating significant financial exposure

Accenture, Data Analyst I | India

Jan 2018 – Mar 2020

- Built ARIMA-based seasonal forecasting models on 250k+ records, improving forecast accuracy by 12% and increasing market base and gross margin by 8%
- Developed self-service Tableau dashboards standardizing engagement KPIs, reducing manual reporting by 25%
- Integrated link analysis and XGBoost into healthcare product efficacy models, improving treatment outcome accuracy and reducing false positives by 20%
- Identified cardiovascular biomarkers using Python statistical analysis and Tableau visualizations, supporting improved healthcare product assessments

ACADEMIC PROJECTS

Strategic Analysis of Hospitality Data | EDA, Classification, Time Series Analysis

- Applied SVM, Logistic Regression, and XGBoost on Expedia hotel data with ARIMA-based trend analysis to model ADR and cancellation behavior

Healthcare Bot | LLM, CUDA

- Built an LLM-powered healthcare assistant using FLAN-T5, GPT-2, LSTM, and BERT, integrating user feedback to improve intent recognition and diagnostic response quality

SKILLS

Programming Languages & Frameworks: Python (NumPy, Pandas, Scikit-Learn), R, PySpark, Java (REST APIs), Git, Bitbucket, JIRA, Linux / UNIX

Databases & Cloud Technologies: SQL, PostgreSQL, MySQL, MongoDB, Apache Spark, AWS EC2, AWS S3, Hadoop, Redshift

Data Analytics & Reporting: Predictive analytics, ETL, feature engineering, supervised and unsupervised ML, NLP, probability & statistics, hypothesis testing, time-series forecasting (ARIMA), Alteryx, Databricks

Data Visualization & Tools: Tableau, Excel, Matplotlib, Seaborn, Power BI, Databricks

EDUCATION

Master of Science, MS in Data Science

Sep 2021 – Dec 2023

Northeastern University, Boston, MA